ClipSAT : Cambridge & Edexcel Additional Mathematics 0606/4PM1

Unit 3.01: Indices, Surds & Polynomials

Name: _____

Q1. The variables x and y are related by the equation y = 2x^3 - 5x. Given that x is increasing at a constant rate of 3text{ units per second}, find the rate of change of y with respect to time at the instant when x = 2.

[ ] A: 57text{ units per second}
[ ] B: 19text{ units per second}
[ ] C: frac{19}{3}text{ units per second}
[ ] D: 21text{ units per second}

Q2. Consider the cubic polynomial fx = x^3 - 5x^2 + 10x - 20 . Find the remainder f2 upon division by x - 2 .

[ ] A: -10
[ ] B: -7
[ ] C: -12
[ ] D: -22

Q3. The diagram shows part of the curve y = 6x - x^2 , which intersects the x -axis at 0, 0 and 6, 0 . Find the area of the region enclosed between the curve and the x -axis.

[ ] A: 27
[ ] B: 36
[ ] C: 54
[ ] D: 72

Q4. Solve the logarithmic equation log_25x + 6 - log_2x = 3 .

[ ] A: x = 2
[ ] B: x = 6
[ ] C: x = frac{1}{2}
[ ] D: x = -3

Q5. Find the value of the integer n ge 3 that satisfies the equation {}^{n}mathrm{P}_{3} = 10 times {}^{n}mathrm{C}_{2}

[ ] A: n = 7
[ ] B: n = 12
[ ] C: n = 5
[ ] D: n = 8

Q6. Find the remainder when the polynomial fx = 4x^3 - 6x^2 + 3x - 1 is divided by 2x - 1 .

[ ] A: -frac{1}{2}
[ ] B: frac{1}{2}
[ ] C: -frac{9}{2}
[ ] D: 0

Q7. In triangle ABC , sides of lengths 32 and 10 meet at an included angle of 60^circ opposite side x . Find the squared length f = x^2 by the Law of Cosines.

[ ] A: 804
[ ] B: 803
[ ] C: 820
[ ] D: 809

Q8. The position vectors of points A and B relative to an origin O are vec{OA} = 2mathbf{i} - 3mathbf{j} and vec{OB} = 7mathbf{i} + 9mathbf{j}. Find the unit vector in the direction of vec{AB}.

[ ] A: frac{1}{13}5mathbf{i} + 12mathbf{j}

[ ] B: frac{1}{13}-5mathbf{i} - 12mathbf{j}

[ ] C: frac{1}{sqrt{61}}5mathbf{i} + 6mathb

[ ] D: frac{1}{15}9mathbf{i} + 12mathbf{j}